

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular End Semester Examination – Summer 2022****Course: B. Tech.****Branch : Civil Engineering****Semester : VI****Subject Code & Name: BTCVC604 Project Management****Max Marks: 60****Date: 23/08/2022****Duration: 3.45 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks**Q.1 Solve Any Two of the following.**

- A) Explain phases in a construction project management. **(CO1) 6**
- B) Draw the network and find the project duration and critical path. **(CO2) 6**

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	--	--	--	A	B	B	C	D	E	G, F	H, I
Duration (Days)	7	10	5	8	10	13	6	5	6	9	7

- C) Find Total float associated with the project in Q. 1 (B) **(CO2) 6**

Q.2 Solve Any Two of the following.

- A) Define network compression and Explain various methods. **(CO3) 6**
- B) Draw network and find the project duration (days) and project cost. Take indirect cost as 100 Rs./day. **(CO3) 6**

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	--	--	--	A	B	B	C	D	E	G, F	H, I
Normal Time	10	15	8	10	15	18	10	12	8	10	10
Normal Cost	8000	10000	5000	7000	15000	20000	3000	4000	7000	5000	2000
Crash time	12	18	10	13	18	21	13	15	10	10	12
Crash cost	10000	11500	7000	8500	15900	21500	3600	4900	7200	5000	2100

- C) Crash the project mentioned in Q.2 B by 5 days and find the crash cost. **(CO3) 6**

Q.3 Solve Any Two of the following.

- A) Differentiate between resource leveling and resource smoothing. **(CO3) 6**
- B) Draw network diagram and find the project duration (days) and critical path. **(CO3) 6**

Activity	1-2	2-3	2-4	3-4	3-5	2-6	4-6	5-6	6-7
to	6	3	2	4	1	5	7	1	3
tm	9	4	5	6	2	6	8	2	5
tp	12	11	14	8	3	7	15	3	7

- C) Find the Z value if the Project mentioned in Q.3 B. has to be completed in 30 days. (CO3) 6

Q.4 Solve Any Two of the following.

- A) Explain various interest formulae and their significance. (CO4) 6
B) Find the value of Rs. 30000 deposited annually after 5 years with 8% p.a. (CO4) 6
C) Fixed cost is 30 lakhs, Variable cost is Rs. 10, If the revenue for each item is Rs. 70, Calculate Break even Quantity. (CO4) 6

Q.5 Solve Any Two of the following.

- A) Explain the process of quality control in construction projects. (CO5) 6
B) Discuss Philosophy of Deming. (CO5) 6
C) Explain use of Computers in Project Management. (CO5) 6

*** End ***

Sub. Que.	Model Answer	Marks	Total Marks
A.	Explain phases in construction project Management → A project consists of four phases 1) Conceptualization 2) Planning 3) Execution 4) Termination 1) Conceptualization phase :- - Starting with an idea - Project idea is conceptualized with initial considerations of all possible alternatives for achieving project objectives - Proposal is developed & estimated cost is found out. - Dimensions are analysed. 2) Planning phase :- - In this stage project structure is planned based on project appraisal & approvals. - Detailed plan for activity, finance & resources are developed. - Major tasks in planning - Identification of activities & sequence - Time frame for execution. - Estimating & budgeting - Staffing. 3) Execution phase :- - In this phase plans are put into operation. - Each activity is monitored, controlled & coordinated to achieve project objective. - Important activities in Execution • Communicating with stakeholders • Receiving progress • Monitoring cost & time • Controlling quality		

Que No.	Sub. Que.	Model Answer	Marks	Total Marks
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• Managing changes.

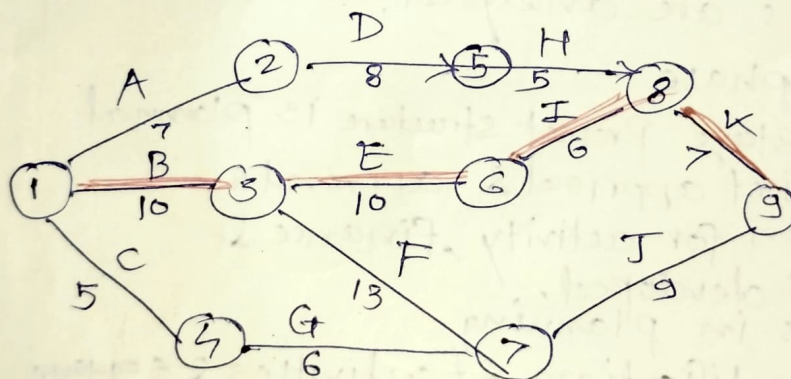
11) Termination phase :-

- this phase marks the completion of project.

- in this phase the agreed deliverables are installed & project put into operations with agreement's to followup & evaluation.

Draw the network & find the project duration & Critical path.

Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	-	-	A	B	B	C	D	E	G, F	H, I
Duration (Days)	7	10	5	8	10	13	6	5	6	9	7



$$A-D-H-K = 7+8+5+7 = 27$$

$$B-E-I-K = 10+10+6+7 = 33 \checkmark$$

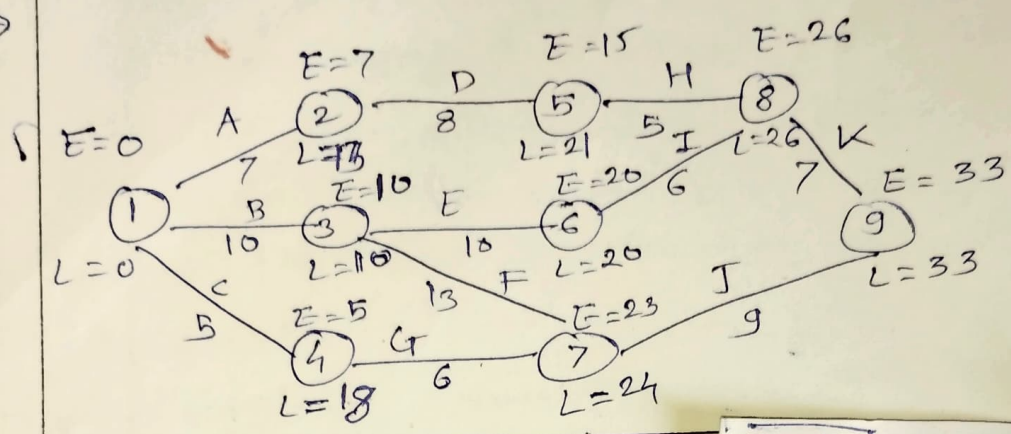
$$B-F-J = 10+13+9 = 32$$

$$C-G-J = 5+6+9 = 20$$

critical path is B-E-I-K

$$\text{Total duration} = \underline{\underline{33}} \text{ days}$$

1 C. Find Total float associated with the project in Q1.B



Activity	Duration	EST	EFT	LST	LFT	Total float
A	7	0	7	0	7	0
B	10	0	10	0	10	0
C	5	0	5	12	17	13
D	8	7	15	13	21	6
E	10	10	20	10	20	0
F	13	10	23	11	24	1
G	6	5	11	18	24	13
H	5	15	20	21	26	6
I	6	20	26	20	26	0
J	9	23	32	24	33	1
K	7	26	33	26	33	0

Total Float
 $= LF - D - ES$

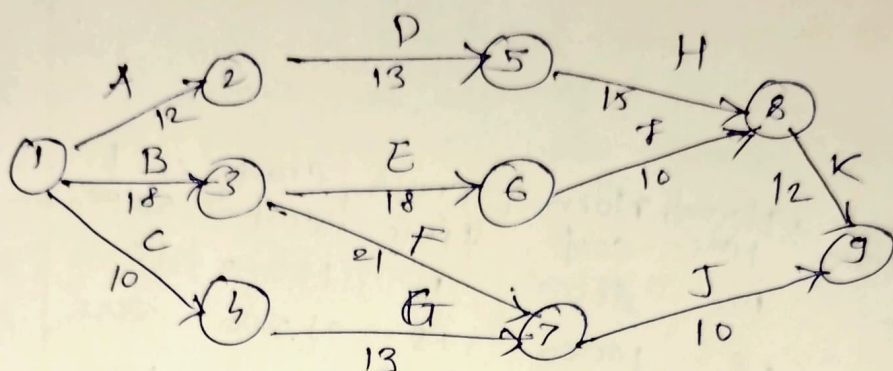
Que No.	Sub. Que.	Model Answer	Marks	Total Marks
2.	A.	<u>Define Network Compression & Explain various Methods</u> - Network Compression. - A project involves completion of No. of activities. - the cost of project depends on activities and change in duration hence cost of activity affects the project duration and total cost of project. - the time cost optimization determines duration & corresponding costs. - the objective of time cost optimization determines normal duration of project. - Network Compression can be done by two methods. ① Crashing ② fast tracking. ① Crashing :- Crashing is a compression techniques and it concentrates on project schedule and gives effort to accelerate the project's completion date. e.g ① Overtime ② Allocating additional resources to specific activity ③ Hiring additional resources ④ Incentive payments for early completion.		

Que No.	Sub. Que.	Model Answer	Marks																																																																								
		2] fast tracking - fast tracking is scheduled compression technique where activities are rearranged to run in parallel which were originally planned in sequence. - when you need to reduce total time of completion of project fast tracking technique can be applied. - in other words fast tracking is process of reducing no. of sequential relationship and replacing them with parallel relationship. - fast tracking can only be applied if the activities in question can actually be overlapped.																																																																									
2 B.		Draw Network and find project duration (days) & project cost. Take Indirect cost as 100 Rs/Day <table border="1"> <thead> <tr> <th>Activity</th> <th>A</th> <th>B</th> <th>C</th> <th>D</th> <th>E</th> <th>F</th> <th>G</th> <th>H</th> <th>I</th> <th>J</th> <th>K</th> </tr> </thead> <tbody> <tr> <td>Predecessor</td> <td>-</td> <td>-</td> <td>-</td> <td>A</td> <td>B</td> <td>B</td> <td>C</td> <td>D</td> <td>E</td> <td>G, F</td> <td>H, I</td> </tr> <tr> <td>Normal time</td> <td>12</td> <td>18</td> <td>10</td> <td>13</td> <td>18</td> <td>21</td> <td>13</td> <td>15</td> <td>10</td> <td>10</td> <td>12</td> </tr> <tr> <td>Normal cost</td> <td>8000</td> <td>10000</td> <td>5000</td> <td>7000</td> <td>15000</td> <td>20000</td> <td>3000</td> <td>4000</td> <td>7000</td> <td>5000</td> <td>2000</td> </tr> <tr> <td>Crash time</td> <td>10</td> <td>15</td> <td>8</td> <td>10</td> <td>15</td> <td>18</td> <td>10</td> <td>10</td> <td>8</td> <td>10</td> <td>10</td> </tr> <tr> <td>Crash Cost</td> <td>10000</td> <td>11500</td> <td>7000</td> <td>8500</td> <td>15900</td> <td>21500</td> <td>3600</td> <td>4900</td> <td>7200</td> <td>5000</td> <td>2100</td> </tr> </tbody> </table>	Activity	A	B	C	D	E	F	G	H	I	J	K	Predecessor	-	-	-	A	B	B	C	D	E	G, F	H, I	Normal time	12	18	10	13	18	21	13	15	10	10	12	Normal cost	8000	10000	5000	7000	15000	20000	3000	4000	7000	5000	2000	Crash time	10	15	8	10	15	18	10	10	8	10	10	Crash Cost	10000	11500	7000	8500	15900	21500	3600	4900	7200	5000	2100	
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Que .No.	Sub. Que.	Model Answer	Marks	Total Marks
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Network



$$A-D-H-K = 52$$

$$B-E-F-K = 58 *$$

$$B-F-J = 49$$

$$C-G-J = 33$$

$\therefore B-E-F-K$ is the critical path

Duration is 58 days

$$\text{Total cost} = \text{Direct cost} + \text{Indirect cost}$$

$$= 86000 + (58 \times 100)$$

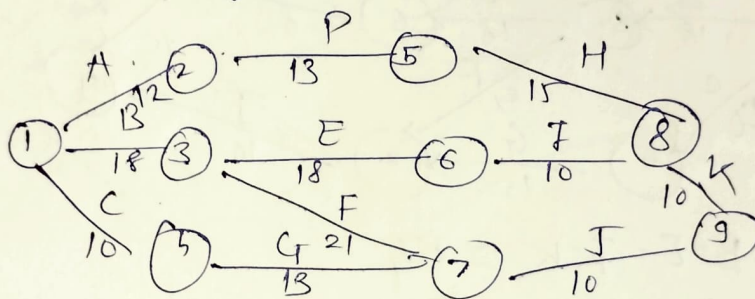
$$\boxed{\text{Total cost} = 91800}$$

Que No.	Sub. Que.	Model Answer	Marks	Total Marks
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Q.2 C Crash the project mentioned in Q.2.B by 5 days:
 & find crash cost.

Critical path	Critical activity	Crash limit	Cost slope
B-E-F-K	B	3	500
	E	3	300
	F	2	100
	K	2	50 *

Crash activity 'K' by 2 days



$$B-E-F-K = 58$$

$$A-D-H-K = 50$$

$$B-F-J = 49$$

$$C-G-J = 33$$

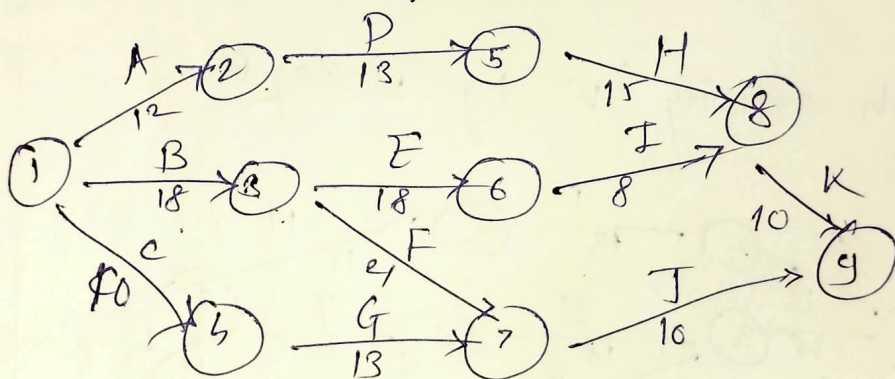
$$\begin{aligned} \text{Total Cost} &= \text{Previous Cost} + \text{Direct Cost of cost slope} - \text{Indirect Cost} \\ &= 91800 + 50 - 200 \end{aligned}$$

$$\boxed{\text{Total Cost} = 91650}$$

Que No.	Sub. Que.	Model Answer	Marks	Total Mark
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Critical path	Critical activity	Crash limit	Cost slope
B-E-I-K	B	3	500
	E	3	300
	I	2	100 *
	K	0	50

Crash activity 'I' by 2 days



$$B-E-I-K = 54$$

$$A-D-H-K = 50$$

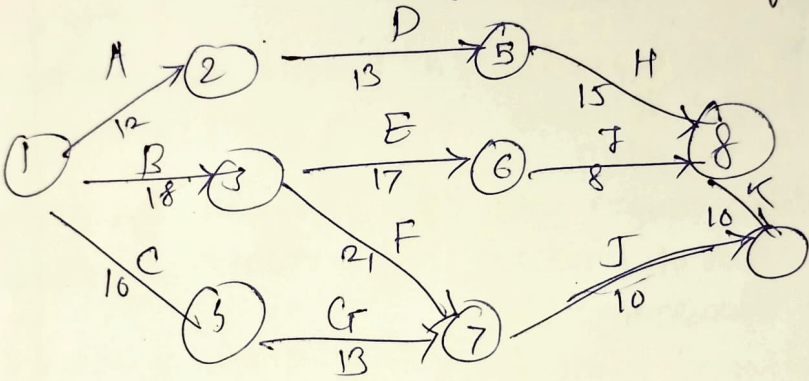
$$B-F-J = 49$$

$$C-G-J = 33$$

$$\text{Total cost} = \text{Previous Cost} + \text{Direct Cost} - \text{Indirect Cost}$$

$$= 91650 + 100 - 200$$

$$\boxed{\text{Total cost} = 91550}$$

Que. No.	Sub. Que.	Model Answer			Marks	Total Marks
		Critical path	Critical activity	Crash limit	Cost slope	
		B-E-J-K	B	3	500	
			E	2	300 *	
			F	0	100	
			K	0	0	
		Crash activity E by 1 day				
						
		$B-E-J-K = 53$ $A-D-H-K = 50$ $B-F-J = 49$ $C-G-J = 33$				
		$\text{Total cost} = \text{Previous cost} + \text{Direct cost} - \text{Indirect cost}$ $= 91530 + 300 - 100$				
		$\text{Total Cost} = 91750$				

Que. No.	Sub. Que.	Model Answer	Marks	Total Marks		
3	A.	<u>Differentiate between resource levelling & resource smoothing.</u> <table><tr><td> <u>Resource levelling</u> 1) Resource limited scheduling technique 2) Project end date may change 3) Can be done to activities on critical path 4) Changes are allowed to critical path 5) May not require additional resources 6) Resource levelling is usually scheduled first </td><td> <u>Resource smoothing</u> 1) Time limited scheduling technique 2) Project end date will not change 3) can not be done to activities on critical path. 4) changes are not allowed to critical path 5) May require additional resources to address left over resource conflicts 6) Resource smoothing is often performed after resource levelling. </td></tr></table>	<u>Resource levelling</u> 1) Resource limited scheduling technique 2) Project end date may change 3) Can be done to activities on critical path 4) Changes are allowed to critical path 5) May not require additional resources 6) Resource levelling is usually scheduled first	<u>Resource smoothing</u> 1) Time limited scheduling technique 2) Project end date will not change 3) can not be done to activities on critical path. 4) changes are not allowed to critical path 5) May require additional resources to address left over resource conflicts 6) Resource smoothing is often performed after resource levelling.		
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Que .No.	Sub. Que.	Model Answer	Marks	Total Mark
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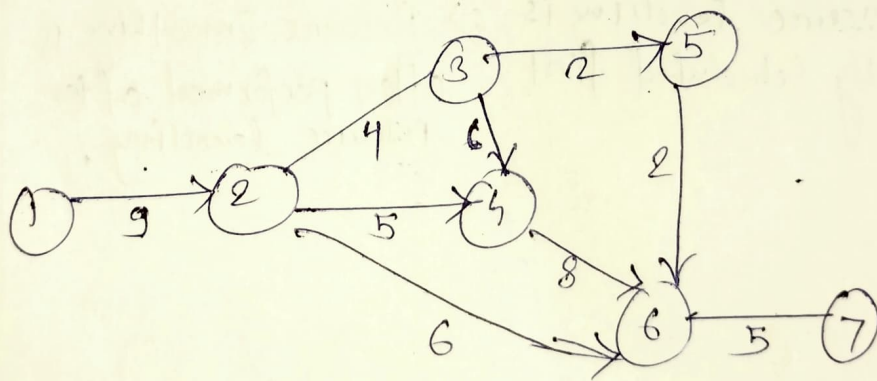
Q.3 B. Draw Network dia. & find project duration (days) & critical path.

Activity 1-2 1-3 1-4 2-3 2-4 3-5 4-5 5-6 6-7

to 6 3 2 4 1 5 7 1 3

tm 9 4 5 6 2 6 8 2 5

tp. 12 11 14 8 3 7 15 3 7



$$1-2-3-5-6-7 = 22$$

$$1-2-3-4-6-7 = 32 *$$

$$1-2-4-6-7 = 27$$

$$1-2-6-7 = 20$$

Que .No.	Sub. Que.	Model Answer	Marks	Total Mark
Q.3	C	Final Z. value if Project Mentioned in Q3 complete in 30 days Total duration = 34 days find Z. if duration is 30 days - sum of var variance along critical path = 1-2-3-4-6-7 $= 1 + 1.77 + 0.44 + 1.77 + 0.44$ $= \underline{\underline{5.44}}$ $Z = \frac{\text{Scheduled day} - \text{Expected day}}{\sqrt{\text{sum of variance}}}$ $= \frac{30 - 34}{\sqrt{5.44}}$ $Z = \frac{-2}{2.33}$ $Z = -0.858$		

Que. No.	Sub. Que.	Model Answer	Marks	Total Marks
24.	A	Solve Explain various Interest formulae & their significance. → the fee paid to the lender for lending a loan is called the interest. - this extra amount or interest is what needs to be paid along with actual loan. - the interest formula includes two types of Interest's ① Simple Interest ② Compound Interest ① Simple Interest :- simple interest is calculated with following formula. $S.I = P \times R \times T$ where P = Principle amount R = Rate of Interest T = Time Amount = Principle + Simple Interest $A = P + S.I$ $A = P + PRT$ $A = P(1 + RT)$		

Que .No.	Sub. Que.	Model Answer	Marks	Total Mar
		II] compound interest is calculated after calculating the total amount over a period of time based on rate of interest and initial principal. - the formula to calculate the compound interest is $C.I = P\left(1 + \frac{r}{n}\right)^{nt} - P$ where P = Principle amount r = rate of interest n = No. of compounding periods per year t = No. of years $A = P\left(1 + \frac{r}{n}\right)^{nt}$		

Que No.	Sub. Que.	Model Answer	Marks	Total Marks
2.4	B	Find value of Rs 30000 deposited annually after 5 years with 8% p.a. → Given $P = 30,000$ $T = 5 \text{ years}$ $R = 8\% =$ $A = P \left(1 + \frac{R}{100} \right)^T$ $= 30000 \times \left(1 + \frac{8}{100} \right)^5$ $= 30000 \times (1.08)^5$ $= 30000 \times 1.469$ $A = 44079$		

Que No.	Sub. Que.	Model Answer	Marks	Total Marks
Q4.	C.	Fixed cost is 30 lakhs, variable cost is Rs 10, If the revenue for each item is Rs 70, Calculate Break even quantity. → Given Fixed cost = 3000000 variable cost = 10 Revenue per unit = 70 $\text{Break even quantity} = \frac{\text{Fixed Cost}}{\text{Revenue per unit} - \text{variable Cost}}$ $= \frac{3000000}{70 - 10}$ BEP = 50,000 units		

Que. No.	Sub. Que.	Model Answer	Marks	Total Marks
25	A.	<u>Explain process of quality control in construction project's.</u> → Quality in construction is defined as meeting or completing the requirements of client or owner. - Quality in construction is employed with confirmity with which specifications are met. - Designer specifies the grade of concrete to be used & Contractor has to use the ingredient's of concrete such that designers desired grade of concrete is obtained. - Quality in construction is related to 1) Satisfying specification in contract 2) Completing project on time. 3) Fulfiling owners requirements within budget. 4) Avoiding disputes. 5) Ensuring facilities performs it's intended purpose. - Poor quality control may lead to structural failures arising from cracks, deformity, Corrosion etc. - there are plenty of factors that affecting quality of construction such as design, materials, machinery, topography, geology, construction technology, etc. - Quality control (QC) in construction is the process of verifying that the project is built to plan, that the tolerance allowable by industry standards & engineering practices have been met & that finished project meet's with quality standards of architect, owner, Engineer, & Contractor.		

Que .No.	Sub. Que.	Model Answer	Marks	Total Marks
Q5	B	Discuss philosophy of Deming. → B. W. Edward Deming was educated as electrical engineer but also did work as a statistician, professor, author, lecturer & management consultant. — Deming became interested in how statistical analysis could be used to achieve better quality control in 1930. * Concepts associated with Deming — 14 points or principles — 7 deadly diseases — Deming cycles (plan-do-check-act) — System of profound knowledge. * 14 points or principles by Deming 1) Create a constant purpose towards 2) Adopt new. 3) mass inspection 4) awarding business on price tag. 5) improve system of production & service 6) modern methods of training 7) supervision & leadership 8) Drive out fear 9) break down barriers 10) Eliminate numerical goals 11) Eliminate work standards 12) Remove barriers to pride of workmanship 13) vigorous programme of education & training for everyone 14) Create a structure that will help push every day on above 13 points.		

Que No.	Sub. Que.	Model Answer	Marks	Total Marks
Q.5	C.	Explain use of Computer in Project Management. - Computer aided project Management gives increasing potential to improve project Management. - Now a days different softwares are used for project management. 1) Microsoft Project :- - it is the worlds most popular project management software developed by Microsoft. - this application helps project Manager to developing plans, assigning resources to task, tracing progress of work, managing budgets & analyzing work load. - Microsoft: project creat's Critical path schedule - Microsoft project has been extended with office project Server & microsoft project web access. - Project Server stores project data in central database. - Microsoft Recognize different classes of users. - Calenders, videos, tables, filters & fields are sorted in global database. * MS project helps in 1) visualise your project plan in standard format 2) schedule tasks & resources consistently & effectively 3) track information about work, duration & resource requirements for your project.		

Que .No.	Sub. Que.	Model Answer	Marks	Total Marks
		4) Generate report's to share in progress meetings. II] Primavera :- - Primavera is enterprise project portfolio management software. - it includes project management, project Management Collaboration & Control capabilities. - it also integrates with other enterprise Softwares such as Oracle ERP & SAP's ERP system. - Primavera was launched by Primavera System in 1983. acquired by Oracle Corporation in 2008. - Using Primavera organization can manage time, tasks, cost's, resources, contract's, change & risks to execute profitable projects. - Primavera is leading project & program management solution for project's of any size. * Importance of Primavera in construction - it keep project on schedule & within budget. - keep project team on top by allowing them in relation to project completion. - Allows Owner's to be informed on project status throughout construction process. - Reduce stress of managing multiple project's at once.		